Rather Ab. Hamid

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Education

Universität Bremen Present
Master of Science in Space Engineering. *Bremen*

- **Relevant Coursework:** Space Propulsion Systems, Structural Design and Analysis, Introduction to Machine Learning and Data Analysis, On-Board Data Handling, Trajectory Optimization, Design of Space Vehicles

Universität Duisburg-Essen 07/2022 - 12/2022
German Language student *Duisburg*

- **Relevant Coursework:** German A1–B1

SRM University May 2016 - June 2020
Bachelor of Technology in Aerospace Engineering. *Chennai, India*

- **Relevant Coursework:** Computational fluid dynamics, Computer aided design, Aerodynamics, Rocket propulsion, Python, Probability & Statistics

Experience

Faserinstitut Bremen July 2023 – Present
Werkstudent: Design, Simulation & Data Analysis *Bremen, Germany*

- Developing and automating data analysis pipelines in Python (Pandas, Matplotlib) to process and visualize experimental data from materials science research.
- My work supports high-profile projects like HyTank (an Airbus project) and WaGuPa, focusing on materials for cryogenic liquid hydrogen (LH2) storage.
- Designed, built, and commissioned a high-pressure helium permeation test stand, integrating all hardware, sensors, and DAQ systems from the ground up.
- Conducting cryogenic material characterization for the HyTank project, including tensile tests on composites submerged in liquid nitrogen ($-196\,^{\circ}\mathrm{C}$).
- Performing coupled Ansys (CFD) simulations for material analysis in composite tank structures.

Resiliente Energiesysteme University of Bremen Aug 2024 – March 2025
Studentische hilfskraft: Mathematical modelling & simulation *Bremen, Germany*

- Developed and implemented mathematical models in Python in Python to simulate and analyze complex, resilient energy systems for the HyBit project.
- Focused on the modeling and simulation of electrochemical systems, such as hydrogen electrolyzers and fuel cells, to understand their behavior within a larger grid.
- Conducted energy system analysis by testing the models against four different power input data sets representing various renewable electricity supply scenarios.
- My work directly supported the techno-economic assessment of these hydrogen systems.
- Utilized libraries like Pyomo, SciPy, Pandas, etc to build and run the simulations.

Polaris Spaceplanes Aug 2023 – Feb 2024
Werkstudent: Flight Demonstrator Assembly & Component Testing *Bremen, Germany*

- CAD modelling, assembly, integration and testing of flight demonstrator parts
- Development of Python post-processing libraries and data infrastructure
- Core team member for the MIRA II, the world's first aerospike rocket plane, supporting its assembly from the ground up.
- Integrated all major vehicle systems, including structural components, avionics, and the AS-1F aerospike rocket engine.
- Co-built and integrated the LOX/Kerosene engine test stand and actively participated in the aerospike hot-fire test campaigns.

- Designed and 3D printed flight components, and was responsible for soldering and processing the entire vehicle's cable harnesses.
- Analyzed all flight and engine test data using Python to validate performance

Rocket Factory Augsburg Jan 2023 – Apr 2023
Test data intern: Rocket Propulsion Test Data Intern *Augsburg, Germany*

- Rocket engine hot-fire and cold-flow test data analysis using Python (numpy, scipy, pandas)
- Visualization using matplotlib and plotly
- Automatic detection of test parameters and spectral analysis
- Documentation of test results
- Used test data to develop and improve Python models for engine performance estimation and behavior prediction.
- Automated a key in-house test procedure using Python, which was "previously processed manually". I developed the script for automatic parameter detection in pressure drop tests.
- Interpreted data and provided active, actionable advice to test engineers after analyzing hot-fire and cold-flow tests.
- Performed spectral analysis (FFT, PSD, spectrograms) on high-frequency data from vibration sensors.
- Supported the propulsion and operations team in all pre- and post-test activities.

SpanTrik Oct 2018 – Jun 2020
Propulsion & Design Intern *Gujrat, India*

- Design, manufacture and analysis of a sounding rocket (RAVEN)
- Propulsion system testing and documentation
- Worked in cross-functional team under project deadlines

Advanced Computational Laboratory, SRM University Oct 2018 – Jun 2020
Student Researcher (Werkstudent) *Chennai, India*

- Led the design and construction of a mini jet impingement wind tunnel from the ground up for experimental research.
- Used the tunnel to conduct flow visualization studies and analyze fluid dynamics.
- Performed Ansys simulations of turbulent flows to compare computational models against experimental data.
- Handled the full experimental data pipeline, data extraction, numerical calculation, and data visualization.
- Designed CAD models for composite parts and fabricated 3D printed test prototypes
- CAD modelling and Ansys simulations
- Experimental work and documentation
- Collaboration with manufacturing for prototyping and testing

Skills

Languages and Tools: Python, Html, CSS, JavaScript, SQL, PowerBI, Tableau, Data Visualization, Git, Microsoft Office, WordPress, Docker, Azure, API's and Unit Testing.
Softwares: Catia, SolidWorks, Siemens NX, Ansys, Abaqus, AutoCAD, LateX.
Libraries: Numpy, Pandas, Tsfresh, Keras and TensorFlow, Scikit-Gstat, Scikit-Learn, Matpotlib, Statsmodels, Seaborn, Open CV, XGBoost, LightGBM, MLFlow, Airflow, Obscy, Prophet, Dask, SciPy, NLTK, SpaCy, Streamlit and PyTorch.

Projects

Global COVID-19 Data Exploration & Trend Analysis
SQL Server, T-SQL, Window Functions, CTEs, Data Visualization Prep

- Conducted advanced Exploratory Data Analysis (EDA) on global pandemic datasets using SQL Server to calculate critical public health metrics, including infection density and mortality risk ratios across different continents.
- Implemented complex SQL logic using Common Table Expressions (CTEs), Temporary Tables, and Window Functions to derive rolling vaccination metrics relative to population growth.
- Engineered optimized database Views to store aggregated findings, facilitating seamless integration with visualization tools like Tableau for identifying regional hotspots and peak infection rates.

Development of a Real-Time Financial Reporting Dashboard for Credit Card Operations
Power BI, SQL, KPI Reporting, Data Modeling, Data Quality Management

- Developed and visualized an interactive Power BI dashboard to control project key figures, providing real-time monitoring of $57M in revenue and enabling data-driven internal management reporting.
- Automated data preparation and reporting processes by orchestrating SQL queries and defining complex DAX measures, ensuring data consistency and plausibility across 46M+ transactions.
- Conducted data quality management and validation to identify specific risk segments, revealing a 6.06% delinquency rate and optimizing the analysis of customer data from various source systems.

Development and validation of a model for the simulation of cryogenically thermally induced microcracks in CFRP
Microcracks, cryogenic testing, Abaqus, FEA, Composite laminates

- Developed a micromechanical FEA model in Abaqus to predict thermally induced microcracking caused by Coefficient of Thermal Expansion (CTE) mismatch after cryogenic cycling.
- Fabricated CFRP laminates with varying Fiber Volume Fractions (FVF) using autoclave curing to generate physical validation data.
- Conducted cryogenic cycling and quantified microcrack density using optical microscopy, achieving a high correlation with simulation predictions.

Development of Cryogenic Hydrogen Tanks for Aviation - HyTank | *Design, Simulation, Testing, Integration*

- Investigating the thermo-mechanical behavior of CFRP-to-CFRP bonded joints under liquid hydrogen (LH2) conditions to ensure structural integrity for zero-emission aircraft.
- Developing customized cryogenic testing methodologies to characterize adhesive performance and fracture mechanics at cryogenic temperatures.
- Designed and operated a high-pressure permeation test rig to quantify leak rates of different specimens.

Fluid Dynamics & Flow Visualization Platform Development - 3 Projects
Experimental Fluid Dynamics, Fabrication, Flow Visualization, MATLAB Data Analysis, Jet Impingement, Vortex Dynamics

- Designed and constructed a custom Mini Jet Impingement Wind Tunnel over a 7-month period, demonstrating end-to-end prototyping and system integration skills using cost-effective commercial components.
- Executed advanced flow visualization campaigns, utilizing a departmental High-Speed Camera to meticulously analyze jet flow dynamics on a flat plate and flow structures issuing from nozzles.
- Performed all post-processing and quantitative analysis of visualization data using MATLAB, extracting key flow features and generating velocity profiles.

Publications

- A. H. Rather, B. T. Kannan, *Design and Development of Impinging Jet Facility for Flow Visualization Studies*, ICAME 2020.
- Nikhil Gaur, A. H. Rather, B. T. Kannan, *Feasibility Study of Ion Generator for Flow Control*, FAME 2020.
- Nikhil Gaur, A. H. Rather, Deepali, B. T. Kannan, *Visualization of Flow Control using Ion Wind on Impinging Jet*.

Bremen, November 20, 2025 Rather Abdul Hamid